

100 Days of Linux - Day 058

Title: Understanding Environment Variables

Today's Goal

Learn what environment variables are, how to view them, create temporary variables, and access their values.

Why This Matters

Environment variables store configuration such as PATH, HOME, API keys, and application settings. Python applications, Git, Docker, Spark, and cloud tools rely heavily on them.

Commands of the Day: env, printenv, echo

Common Syntax:

```
env
printenv
printenv HOME
echo $HOME
export CITY=Bangalore
echo $CITY
```

Common Variables:

HOME, PATH, USER, SHELL, PWD

Beginner Lesson

An environment variable is a name that stores a value. Programs read these values to know where files are located, which user is logged in, or how they should behave. Variables created with **export** last only for the current terminal session unless permanently added to a shell configuration file.

Practice Questions

1. Run **env** and observe the environment variables available in your terminal.
2. Use **printenv** to display the values of HOME and USER.
3. Display the values of PATH and SHELL using **echo**.
4. Create a temporary environment variable named COURSE with the value Linux100 using **export**, then display its value.
5. Display your current working directory after completing today's tasks.

Hints

Variables are referenced with a dollar sign, such as \$HOME or \$PATH.

Mini Challenge

Create three temporary variables named PROJECT, DOMAIN, and YEAR. Display all three with echo, then verify PROJECT exists using printenv PROJECT.

Common Mistakes

Forgetting to use the dollar sign when reading a variable or expecting exported variables to remain after closing the terminal.

Did You Know?

The PATH variable tells Linux where to search for executable programs when you type a command.

Pro Tip

When troubleshooting software, checking PATH, HOME, and other environment variables is often one of the first diagnostic steps.

Answer Key (Instructor Only)

Q1: env

Q2: printenv HOME && printenv USER

Q3: echo \$PATH && echo \$SHELL

Q4: export COURSE=Linux100 && echo \$COURSE

Q5: pwd

Homework

Create five temporary environment variables with different values. Open a new terminal and observe which variables disappear.

Next Day Preview

Tomorrow you'll learn aliases to create custom shortcuts for frequently used Linux commands.